

Section A

1. [Maximum mark: 6]

X	0	1	2	3
$P(X=x)$	$\frac{3}{13}$	$\frac{1}{13}$	$\frac{4}{13}$	k

- (a) Find the value of k . [3]
- (b) Find $E(X)$. [3]

[illegible]

3. [Maximum mark: 6]

Consider the function $f(x) = \frac{3x+1}{x-2}$, $x \neq 2$.

(a) For the graph of f ,

(i) write down the equation of the vertical asymptote;

(ii) find the equation of the horizontal asymptote.

[3]

Let $g(x) = x^2 + 4$, $x \in \mathbb{R}$.

(b) Find $(f \circ g)(1)$.

[3]

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3. [Maximum mark: 7]

Let $f(x) = \frac{6x-1}{2x+3}$, for $x \neq -\frac{3}{2}$.

(a) For the graph of f ,

- (i) find the y -intercept;
- (ii) find the equation of the vertical asymptote;
- (iii) find the equation of the horizontal asymptote.

[5]

(b) Hence or otherwise, write down $\lim_{x \rightarrow \infty} \left(\frac{6x-1}{2x+3} \right)$.

[2]

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11. Consider the curve $y = 5x^3 - 3x$.

(a) Find $\frac{dy}{dx}$. [2]

The curve has a tangent at the point $P(-1, -2)$.

(b) Find the gradient of this tangent at point P. [2]

(c) Find the equation of this tangent. Give your answer in the form $y = mx + c$. [2]

Working:

Answers:

- (a)
 (b)
 (c)



3. [Maximum mark: 6]

Consider the function $f(x) = x^2 e^{3x}$, $x \in \mathbb{R}$.

(a) Find $f'(x)$. [4]

(b) The graph of f has a horizontal tangent line at $x = 0$ and at $x = a$. Find a . [2]

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